

A 16-Bit-Native Compression Pipeline for Lossless Medical Images

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Abstract—Medical images must be stored and transmitted losslessly, since a radiologist must see exactly what the scanner produced, and they are “deep,” carrying 10–16 bits per pixel rather than the 8 bits of an ordinary photo. That deep format is where most compression tools perform poorly, because nearly all of them were adapted from photographic and general-purpose imaging, where pixels are 8 bits, rather than designed for medical data. This paper describes MIC (Medical Image Codec), a lossless codec designed natively for medical imaging that works directly on 16-bit data instead of splitting each pixel into two bytes. MIC chains three simple stages, a spatial predictor, a 16-bit run-length encoder, and a large-alphabet entropy coder called Finite State Entropy (FSE, a fast table-driven form of asymmetric numeral systems), and adds a multi-state decoder that keeps the CPU more fully utilized to speed up decompression. On 39 real DICOM images across eleven modalities, MIC compresses about 10% better on average (better on 34 of the 39 images) than a strong general-purpose baseline (Delta + Zstandard), reaches a geometric-mean compression ratio of $3.36\times$ (within about 2% of High-Throughput JPEG 2000 at $3.42\times$ and about 88% of JPEG-LS at $3.84\times$), and is the fastest decoder of the four tested codecs on all 39 images on ARM64 and 36 of 39 on AMD64. A 20 KB pure-JavaScript decoder lets the format be decoded directly in a web browser. The main caveat is that this is a 39-image study; the results are encouraging but need confirmation on larger, multi-hospital datasets.

Index Terms—medical image compression, lossless compression, entropy coding, asymmetric numeral systems (ANS), finite state entropy (FSE), DICOM, high-bit-depth imaging.

I. WHY MEDICAL IMAGES NEED A DIFFERENT CODEC

MEDICAL imaging plays a central role in understanding the human body, and as the instruments that capture these images improve, both their resolution and the volume of imaging data they generate continue to rise. A modern scanner produces a great deal of data: a single digital breast tomosynthesis view can comprise 60 or more image slices and over 600 MB, and a full screening study can exceed 2 GB uncompressed. Hospitals store and move very large volumes of this data every day, so effective compression directly affects storage cost, network load, and how quickly images appear in a viewer. Because a radiologist must see exactly what the scanner produced, this compression must be lossless: the reconstructed image has to match the original bit for bit, with no loss of fidelity.

DICOM [1], the medical imaging standard, already supports several lossless formats—JPEG 2000 [2], HTJ2K [3], JPEG-LS [5], and a basic run-length scheme. Each makes a different

trade-off between how small the file gets, how fast it decodes, and how complex it is to implement [18], [19]. JPEG 2000 and HTJ2K compress well but rely on fairly heavy machinery. JPEG-LS usually gets the smallest files. Plain run-length coding is simple but barely compresses.

Lossless compression for medical images is not new, but a common thread runs through these standards: none of them was designed natively for medical imaging. Their prediction and entropy stages originated in photographic and general-purpose imaging, where pixels are typically 8 bits, and were later applied to the 10–16-bit data that medical scanners produce. MIC takes the opposite approach and is built from the ground up for medical images.

A. The core problem: most coders assume 8-bit data

This is the central observation behind the paper. The widely used entropy coders, including Huffman, arithmetic coding, the LZ77 family, and the FSE/ANS coder inside Zstandard [11], were all designed for **8-bit byte streams**, where there are only 256 possible symbols. Medical images do not fit that assumption: a 16-bit pixel has up to 65,536 possible values.

When an 8-bit coder is applied to 16-bit data, it has two poor options:

- 1) **Split each pixel into a high byte and a low byte** and code them separately. This doubles the number of coding steps and, more importantly, discards the relationship between the two bytes.
- 2) **Cap the alphabet at 256 values** and store anything rarer as raw data, which wastes space.

Why does splitting hurt? Consider a tiny prediction error like -2 , -1 , 0 , or $+1$. The low byte carries the real information, but the high byte is almost always the same (just a sign-extension). A byte-by-byte coder still spends bits encoding that high byte even though it is nearly fully determined by the low one. On the test images, the information shared between the two bytes ranges from 0.3 to 1.1 bits per pixel, which is the ceiling on how much a byte-splitting coder is wasting, and MIC’s measured 5–22% advantage falls within that ceiling.

a) *The residual stream:* Throughout, a *residual* is the difference between a pixel and a prediction of it. Example: if the predictor guesses 1020 and the true pixel is 1023, the residual is $+3$. Smooth regions of an image produce many residuals near zero, and the same small values recur frequently, which is exactly the pattern run-length and entropy coding exploit well.

B. What MIC is

MIC is designed around the characteristics of medical images themselves. Such images tend to contain large homogeneous regions and comparatively few sharp edges or abrupt changes in contrast, and their pixels are commonly 10–16 bits deep. Starting from these properties, we surveyed entropy-coding and signal-processing techniques to identify a combination that compresses this kind of data well while still decoding at high throughput. The result is a three-stage pipeline.

MIC keeps the residuals at their native 16-bit width and combines (i) a simple spatial predictor, (ii) a 16-bit run-length encoder, and (iii) an entropy coder (FSE) extended to handle large alphabets. Plus a multi-state decoder that improves decompression speed by letting the CPU work on several symbols at once.

C. Contributions, briefly

- A **16-bit-native pipeline** (Delta + 16-bit RLE + extended FSE) that outperforms Delta + Zstandard on the large majority of images (34 of 39, by about 10% in geometric mean).
- An **entropy coder for large alphabets**—up to 65,535 symbols, versus the 4,096-symbol limit in Zstandard’s FSE. Tables shrink automatically to fit the data (as small as ~2 KB for 8-bit content).
- A **multi-state decoder** that runs 2/4/8 independent decode chains to use idle CPU capacity, with no cost to compression ratio.
- An **evaluation** against HTJ2K, JPEG-LS, and Delta + Zstandard.
- A **20 KB JavaScript decoder** plus a WebAssembly build, for client-side decoding in browsers.

a) *Scope*: This work covers lossless, single-frame, grayscale images only. Color and multi-frame extensions, lossy modes, and progressive decoding are out of scope.

II. RELATED WORK

The DICOM lossless codecs. JPEG 2000 [2] uses a wavelet transform plus a sophisticated block coder (EBCOT): strong ratios, but slow decoding. HTJ2K [3], [4] replaces the block coder with a faster one and decodes considerably quicker. JPEG-LS [5] uses a small predictor (Median Edge Detector) and Golomb-Rice coding; it is a widely deployed standard and a strong ratio baseline. The basic DICOM run-length scheme compresses little because it works on raw bytes with no prediction step. Research codecs such as CALIC [6] and FLIF [7] compress well on natural images but are not DICOM standards and lack browser decoders.

These are the codecs MIC is measured against. The goal is not to outperform them on every metric, but to explore a **simpler, residual-domain design** aimed at high-bit-depth images, with strong speed and easy deployment.

Entropy coding background. Asymmetric Numeral Systems (ANS) [8], [9] is a family of entropy coders that replaces arithmetic coding’s interval math with simple integer state

updates. FSE is a fast, table-driven version of ANS [10] made popular by Zstandard [11]. The limitation is that these implementations are tuned for byte streams (Zstandard caps FSE at 4,096 symbols).

Parallel ANS. Prior work has shown that ANS decoders run faster when several independent streams are decoded at once (Giesen on interleaved coders [12]; later work scaling this to many streams [13] and to GPUs [14]). MIC does not claim to invent ANS or interleaving; it applies these known ideas inside a 16-bit-native medical codec with a specific large-alphabet design.

Table I places MIC against the prior work across five dimensions relevant to medical image compression.

TABLE I
FEATURE COMPARISON OF MIC WITH RELATED CODECS AND ENTROPY CODING METHODS. “ENTROPY CODER SEES 16-BIT RESIDUAL AS ONE SYMBOL” IS A PROPERTY OF THE ENTROPY STAGE, NOT OF THE CODEC’S OVERALL BIT-DEPTH SUPPORT: JPEG-LS AND HTJ2K BOTH HANDLE 16-BIT MEDICAL IMAGES, BUT THEIR ENTROPY STAGES (GOLOMB-RICE ON CONTEXT BINS FOR JPEG-LS, BIT-PLANE CODING FOR HTJ2K) DO NOT TREAT THE PREDICTED RESIDUAL AS A SINGLE 16-BIT SYMBOL. SOURCE-LINE COUNTS ARE ORDER-OF-MAGNITUDE CONTEXT ONLY; THE REFERENCE LIBRARIES ALSO IMPLEMENT STANDARD PROFILES OUTSIDE MIC’S CURRENT SCOPE.

Feature	JPEG-LS	HTJ2K	Zstd	rANS	MIC
Lossless medical	Yes	Yes	General	General	Yes
Entropy coder sees 16-bit residual as one symbol	No	No	No	Configurable	Yes
Multi-state ILP	No	No	No	Yes	Yes
Browser decoder	Via WASM port	Via WASM port	Partial	No	Purpose-built JS
Source lines	5k	20k	N/A	N/A	1.1k

III. HOW MIC WORKS

The pipeline is three stages, each simple enough to decode in one straight-line pass with very few branches: raw 16-bit pixels go through spatial prediction (subtract a guess based on neighbors), then 16-bit run-length encoding (collapse repeated values), then FSE entropy coding (remove the remaining redundancy), producing compressed bytes.

A. Spatial prediction

For each pixel, MIC predicts its value as the **average of the pixel above and the pixel to the left**, then stores only the difference (the residual). Pixels on the top edge or left edge use whatever single neighbor is available; the very first pixel is stored as-is. Division rounds down, and the decoder repeats the exact same arithmetic so reconstruction is exact.

This predictor was chosen because it is inexpensive, has almost no branches on the decode side, and works well on medical images.

a) *Other predictors we evaluated*: We compared four predictors (left-only, average, Paeth from PNG [16], and MED from JPEG-LS [5]) through the full pipeline. MED gave the best ratio (about 2.4% better than average) but decoded 1.5–2× slower because of its three-way branch and a diagonal dependency that blocks the fast branch-free loop. The average predictor was preferred for its consistency and speed, so it is the default. Table II summarizes the comparison.

TABLE II
PREDICTOR ABLATION SUMMARY: GEOMETRIC MEANS AND WIN COUNTS ACROSS ALL 39 IMAGES.

Predictor	Geo. Mean	Wins	Avg→X
Left-only	3.22×	2/39	−4.4%
Average (MIC)	3.36×	11/39	baseline
Paeth	3.39×	0/39	+0.8%
MED (JPEG-LS)	3.44×	26/39	+2.4%

B. Handling large jumps: overflow coding

Most residuals are small, but occasionally a pixel jumps far from its prediction. Rather than widen the whole stream to 32 bits to handle rare large values, MIC reserves one special code as a **delimiter**: small, in-range residuals are stored directly as 16-bit codes (with zero mapped to a fixed midpoint value); a large, out-of-range residual is stored as the delimiter followed by the raw pixel value.

On decode, the reader checks each value: if it is the delimiter, the next value is a raw pixel; otherwise it is a normal residual. The ranges are designed not to overlap, so there is never any ambiguity. In normal 8–16-bit clinical data the overflow path is rare; it exists to guarantee correctness, not for the common case. Table III shows the mapping for a representative bit depth.

TABLE III
OVERFLOW CODING EXAMPLE FOR $d = 12$ BITS (DELTA THRESHOLD = 2047, DELIMITER $D = 4095$).

Residual r	Emitted code(s)	Notes
−2	2045	in-range
−1	2046	in-range
0	2047	in-range (= threshold)
+1	2048	in-range
+2	2049	in-range
±2047	4095, raw pixel	overflow

C. 16-bit run-length encoding (RLE)

The residual stream is turned into two kinds of runs: **same runs** (a count plus one repeated value, used when a value appears at least three times in a row) and **literal runs** (a count followed by that many values copied verbatim; the values need not be distinct). A single header number tells the two apart using a midpoint trick: small header values mean “same run of this length,” large ones mean “literal run.” The longest single run is ~32,767 symbols; longer stretches are split across several headers automatically.

Why 16-bit RLE matters. Suppose 1,000 identical 16-bit residuals appear in a row. MIC encodes that as just two numbers: a count and the value. A *byte*-oriented RLE sees the high and low bytes interleaved, so no single byte repeats 1,000 times, so it is forced into two interleaved runs or falls back to literals. Working at 16 bits captures the repetition that byte-splitting hides.

Fast decode. Same runs (often 80% of all runs on smooth images) are fast-pathed: the decoder just hands back the cached value and decrements a counter, without touching the compressed data at all.

D. The entropy coder (FSE) for large alphabets

FSE in one paragraph. FSE keeps a single integer “state.” To decode one symbol, it looks up the current state in a table; the entry tells it which symbol to output, how many bits to read next, and how to compute the next state. So each symbol costs **one table lookup, one bit read, and one addition**—no multiplies or divides. (Encoding runs the symbols in reverse and builds the bitstream backwards; decoding reads forwards. That reversal is inherent to how ANS works.)

MIC extends FSE to handle up to **65,535 symbols** instead of Zstandard’s 4,096, because high-bit-depth residuals genuinely require a large alphabet. It also **sizes its tables to the data**: an 8-bit image stored in 16-bit containers (common for MR) shrinks the working tables from ~512 KB down to about 2 KB, which keeps everything in fast cache.

Picking the table size automatically. FSE has a “tableLog” knob that sets how large and precise the probability table is. MIC selects it based on how many distinct values it sees and how much data backs each value: it uses a larger table only when there are many active symbols *and* enough data to estimate their probabilities reliably (for example, mammography images after prediction). On the test set this adaptive choice correctly turned on the larger table for exactly the nine images that benefit (1–10% better ratio), and decode speed is unaffected by the choice. Table IV shows the ablation across selected images.

TABLE IV
TABLELOG ABLATION: FORCED TL=11/12/13 VS. ADAPTIVE SELECTION (SELECTED IMAGES, ARM64 REFERENCE PLATFORM).

Image	TL=11	TL=12	TL=13	Adaptive
CR	3.47×	3.63×	3.69×	3.69×
MG1	8.00×	8.57×	8.79×	8.79×
MG2	7.98×	8.55×	8.77×	8.77×
MR2	3.14×	3.24×	3.28×	3.28×
RG2	3.85×	4.12×	4.23×	4.23×
RG3	5.64×	5.95×	6.08×	6.08×

IV. FASTER DECODING: MULTI-STATE DECODING

A. The bottleneck

A standard single-state ANS decoder has an inherent limitation: each symbol depends on the one before it. To decode symbol N the decoder needs the state produced by symbol $N - 1$, so the work cannot start early. With a table lookup taking roughly 4–5 CPU cycles, the loop is limited to about one symbol every few cycles, even though modern CPUs have 4–6 execution ports that remain largely idle. A single decode chain uses only one of them, leaving ~75–83% of the CPU’s capacity unused.

B. The approach

MIC runs **several independent decoders at once** on interleaved positions. With 8 chains, chain 0 handles positions 0, 8, 16, . . . ; chain 1 handles 1, 9, 17, . . . ; and so on. The chains share nothing during decoding, each has its own state and its own stream of lookups, so the CPU’s out-of-order engine can keep all of them in flight simultaneously.

On the encoder side, the symbols are split into N interleaved groups, each compressed independently, and the resulting streams are concatenated. They all share one decode table, and each chain’s starting state is recorded in the header. The decoder just writes each chain’s output back to its positions.

C. Measured speedup

In theory, N chains could be up to $N\times$ faster. In practice the speedup is smaller because the chains share some bookkeeping (refilling the bit reader), the unrolled loop puts pressure on the instruction cache, and reading the compressed data has its own bandwidth limit [17]. Measured on the FSE stage alone, **4 states** give +36% on ARM64 and +12% on AMD64 (relative to single state), and **8 states** a further +5% on ARM64 and +1% on AMD64, with **no change to the compressed file**. ARM64 continues to improve as chains double; AMD64 flattens out earlier because its core already extracts substantial parallelism from a single chain. Table V shows the simple latency model against the measured ARM64 speedup.

TABLE V
MULTI-STATE DECODER LATENCY MODEL AND EMPIRICAL SPEEDUP.

States (k)	Amortized latency	Theoretical	Empirical (ARM64)
1	L cy/sym	1.0×	1.0×
2	$L/2$ cy/sym	2.0×	1.3×
4	$L/4$ cy/sym	4.0×	1.5×
8	$L/8$ cy/sym	8.0×	1.6×

The decoder signals its mode with a short magic header (a reserved byte value that cannot be a valid single-state setting), so single-, two-, four-, and eight-state files all coexist without confusion. On AMD64 and ARM64 the inner loop uses each architecture’s native variable-shift instructions; other platforms get a pure-Go fallback. Table VI summarises the on-disk layout of a single-frame stream.

TABLE VI
SINGLE-FRAME MIC BITSTREAM LAYOUT. ALL MULTI-BYTE FIELDS ARE LITTLE-ENDIAN. STATES k IS THE NUMBER OF INTERLEAVED FSE CHAINS ($k \in \{1, 2, 4, 8\}$).

Field	Size	Meaning
Magic / mode byte	1 B	Single-state: encodes <code>tableLog</code> (value 5–17). Multi-state: 0xFF.
Multi-state tag	1 B	For multi-state streams, one of 0x02, 0x04, 0x84 indicating $k = 2, 4$, or 8.
Symbol count	4 B	Number of decoded RLE symbols.
TableLog	1 B	Decode-table size: 2^{tableLog} entries (multi-state path only; the single-state path overloads byte 0).
Normalized counts	variable	FSE normalised symbol frequencies, serialised with the standard variable-length encoding.
Per-chain initial states	$k \times 2$ B	Starting state for each FSE chain.
Per-chain stream lengths	$k \times 4$ B	Compressed-stream length per chain (omitted when $k = 1$).
Per-chain compressed streams	variable	Concatenated chain payloads.

V. EVALUATION METHODOLOGY

Dataset. 39 de-identified DICOM images across eleven modalities (CT, MR, CR, X-ray, mammography, digital radiography, nuclear medicine, PET, fluoroscopy, secondary

capture, and tomosynthesis), drawn from the public NEMA WG-04 sample sets [22], the NEMA 1997 CD, a public breast-tomosynthesis case [23], and additional grayscale slices from public GDCM and TCIA collections. It is a varied set but small relative to a real clinical archive, so the results are reported as “on this 39-image dataset” rather than as general claims. Table VII lists the images.

TABLE VII
TEST DATASET: 39 CLINICAL DICOM IMAGES SPANNING ELEVEN MODALITIES.

Image	Modality	Dimensions	Raw (MB)
MR	Brain MRI	256×256	0.13
CT	CT scan	512×512	0.50
CR	Comp. radiography	2140×1760	7.18
XR	X-ray	2048×2577	10.1
MG1	Mammography	2457×1996	9.35
MG2	Mammography	2457×1996	9.35
MG3	Mammography	4774×3064	27.3
MG4	Mammography	4096×3328	26.0
CT1	CT scan	512×512	0.50
CT2	CT scan	512×512	0.50
MG-N	Mammography	3064×4664	27.3
MR1	Brain MRI	512×512	0.50
MR2	Brain MRI	1024×1024	2.00
MR3	Brain MRI	512×512	0.50
MR4	Brain MRI	512×512	0.50
NM1	Nuclear med.	256×1024	0.50
RG1	Radiography	1841×1955	6.86
RG2	Radiography	1760×2140	7.18
RG3	Radiography	1760×1760	5.91
SC1	Sec. capture	2048×2487	9.71
XA1	Fluoroscopy	1024×1024	2.00
CT_BRAIN	CT scan	512×512	0.50
CT_ANKLE	CT scan	512×512	0.50
CT_ORT	CT scan	512×512	0.50
CT_CHEST	CT scan	400×512	0.39
MR_HEAD	Brain MRI	256×256	0.13
MR_INTERA	Brain MRI	1024×1024	2.00
DX_HAND	Digital radiography	1480×1410	3.98
DX_CHEST	Digital radiography	1416×1420	3.84
CR_THORAX	Comp. radiography	2076×1876	7.43
PET1	PET	256×256	0.13
PET2	PET	256×256	0.13
CT_LUNG	CT scan	512×512	0.50
CT_PANCREAS	CT scan	512×512	0.50
MR_BRAIN	Brain MRI	320×256	0.16
MR_BREAST	Breast MRI	256×256	0.13
MR_PROSTATE	Prostate MRI	320×320	0.20
PET_PSMA	PET	200×200	0.08
PET_LUNG	PET	200×200	0.08

Baselines. HTJ2K (via OpenJPH [20]), JPEG-LS (via CharLS [21]), and Delta + Zstandard [11] at its maximum level. All were called in-process (no file or subprocess overhead) for a fair speed comparison.

Metrics. Compression ratio (uncompressed ÷ compressed), and decode and encode throughput in MB/s of uncompressed pixels.

Platforms. Two reference machines: an ARM64 (Apple M4 Pro) and an AMD64 (AWS Intel Xeon 6). Go code used the standard compiler; C components were built with -O3. Each benchmark ran the full image 10 times; run-to-run variance stayed under 2% on ARM64 and 5% on AMD64.

The “MIC” column in the result tables is the fastest available C decoder on each platform: the eight-state decoder, with AVX-512 acceleration of the run-length and prediction-reversal stages on AMD64. Both read identical compressed bytes; only the vector width differs.

Browser decoder. A 20 KB pure-JavaScript ES module (zero npm dependencies) plus a 2.5 MB Go WebAssembly

build. The JS decoder auto-detects 1/2/4/8-state streams and was benchmarked under Node.js on the ARM64 machine.

VI. RESULTS

A. Compression ratio

JPEG-LS produces the smallest files on nearly every image (37 of 39); its context-adaptive predictor is difficult to beat on pure ratio. MIC, however, is not optimizing ratio alone; it targets the best balance of ratio, speed, and simplicity. Across the dataset MIC reaches a **3.36×** geometric-mean ratio: about 88% of JPEG-LS (3.84×) and within about 2% of HTJ2K (3.42×), while decoding faster than both. Mammography compresses best (up to 8.79×) because its smooth tissue produces long near-zero runs, and studies with large uniform backgrounds reach the highest single-image ratios (PET_PSMA at 10.12×, CT_ANKLE at 9.48×). Table VIII gives the per-image ratios for all variants.

B. Encoding speed

MIC encodes faster than the others on most images, on both platforms, because its encoder is a single pass (predict, run-length, table-driven entropy code) while HTJ2K and JPEG-LS do more work per pixel. On **ARM64**, MIC is fastest on 38 of 39 images: roughly 1.0–3.3× HTJ2K, 1.0–7.1× JPEG-LS, and 30–320× Delta + Zstandard (whose max-level LZ77 search accounts for its low throughput); HTJ2K edges ahead only on the tiny PET_LUNG study. On **AMD64**, MIC is fastest on 32 of 39; HTJ2K leads on the two largest mammography images, one radiography image, and four of the smaller new-corpus studies, because the AMD64 baselines are themselves heavily vectorized and per-image table-build overhead weighs more on tiny images. Table IX gives the per-image encoding throughput.

C. Decoding speed

Decoding is where MIC performs best. Its decoder is dominated by the compact FSE loop (one lookup, one bit read, one add) with no hard-to-predict branches. HTJ2K and JPEG-LS both have data-dependent branches that cause CPU mispredictions. Zstandard is the most comparable codec (same predictor, also uses FSE) but works on bytes, so it pays the 2× byte-splitting cost on deep data. On **ARM64**, MIC is fastest on all 39 images: roughly 1.2–2.1× HTJ2K, 1.0–2.0× Zstandard, and 1.1–5.1× JPEG-LS, with the largest margins on the smaller images. On **AMD64**, MIC is fastest on 36 of 39 images; on the three exceptions (CT_BRAIN, MR_INTERA, and the tiny PET_LUNG study) in-process Zstandard is marginally faster. Across its wins MIC runs about 1.0–1.8× HTJ2K and 1.9–7.1× JPEG-LS, and stays within 0.7–1.8× of Zstandard; switching from four to eight states (plus the AVX-512 widening of the post-entropy stages) is what turned several remaining close cases into MIC wins.

MIC therefore delivers both a **better ratio than Zstandard on most images** (34 of 39) and **faster decoding on nearly every image**, which reflects the 16-bit-native design. JPEG-LS retains about a 14% ratio advantage, so the choice between

them comes down to decode speed versus storage cost. Per-image numbers are in Table X, and the entropy stage in isolation is in Table XI.

D. Other MIC variants (for reference)

The main tables use one MIC column, but the codebase includes several variants that exist for different deployment needs rather than to compete on raw speed. The **pure Go** build runs the entire pipeline in Go with no C dependency, at roughly 40–60% of the C decoder’s speed; this is useful where C cannot be linked (mobile, WebAssembly, locked-down runtimes). The **1/2/4-state decoders** are the same C decoder with fewer chains; they serve as baselines for the multi-state comparison. **Wavelet-based MIC** replaces the predictor with a reversible 5/3 wavelet transform [15] feeding the same back end; it is competitive on a few images but generally slower on this corpus, since smooth medical images do not reward the wavelet’s additional cost. The eight-state C decoder is both the simplest and the fastest configuration on this dataset.

E. Multi-threaded decoding

MIC can also split an image into horizontal strips and decode them on separate threads. This scales nearly linearly up to four threads on both platforms and keeps scaling to eight threads on images above ~1 MB, hitting about **4.3 GB/s** on the largest mammography images on ARM64. Small images do not benefit: thread setup costs more than it saves, and on the two 0.08 MB PET images the strips fall back toward raw storage, so strip parallelism does not apply. We report this separately because the other codecs run single-threaded in our pipeline, so a direct comparison would not be fair. Table XII gives the per-image strip-parallel numbers.

Practical guidance: to decode many small images, use single-threaded MIC per request, which is already fast. To decode one large image on a multi-core machine, use strip-parallel mode with four to eight threads.

F. In the browser

The JavaScript decoder performs well. Notably, the **four-state** variant runs 6–21% *faster* than eight-state in JavaScript, even on the same hardware where eight states win in C. The reason is the JavaScript engine (V8): it can keep four independent chains busy but not eight, so the extra chains add overhead without shortening the critical path. The eight-state path still works in JavaScript (it lets one decoder accept C-encoded files without conversion), but four states is the best configuration for the browser.

Concretely, a 9.35 MB mammography image decodes in 17.1 ms (547 MB/s) using four-state strips across worker threads. That means a web DICOM viewer can pull a .mic file straight from object storage and decode it client-side, with no server doing compression or format conversion. (These numbers are from Node.js; full in-browser benchmarking is future work.) Tables XIII and XIV give the JS numbers.

TABLE VIII
LOSSLESS COMPRESSION RATIOS: ALL CODEC VARIANTS, 39 IMAGES. BOLD = BEST RATIO PER ROW.

Image	Raw (MB)	Δ +Zstd-19	MIC	Wavelet	PICS-4	PICS-8	HTJ2K	JPEG-LS
MR	0.13	1.95×	2.35×	2.38×	2.28×	2.21×	2.38×	2.52×
CT	0.50	2.05×	2.24×	1.67×	2.15×	1.96×	1.77×	2.68×
CR	7.18	3.24×	3.69×	3.81×	3.70×	3.71×	3.77×	3.96×
XR	10.1	1.43×	1.74×	1.76×	1.75×	1.75×	1.67×	1.76×
MG1	9.35	7.20×	8.79×	8.66×	8.84×	8.87×	8.25×	8.91×
MG2	9.35	7.21×	8.77×	8.65×	8.83×	8.85×	8.24×	8.90×
MG3	27.3	2.09×	2.24×	2.31×	2.31×	2.33×	2.22×	2.38×
MG4	26.0	3.10×	3.47×	3.59×	3.58×	3.62×	3.51×	3.71×
CT1	0.50	2.47×	2.79×	2.48×	2.54×	2.29×	2.70×	3.19×
CT2	0.50	3.24×	3.48×	2.87×	3.11×	2.72×	3.29×	4.54×
MG-N	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.23×	2.38×
MR1	0.50	1.79×	2.09×	2.14×	2.10×	2.08×	2.12×	2.30×
MR2	2.00	2.77×	3.28×	3.34×	3.31×	3.31×	3.35×	3.52×
MR3	0.50	3.47×	3.92×	4.09×	3.89×	3.83×	4.33×	4.51×
MR4	0.50	3.58×	4.12×	4.18×	4.09×	4.03×	4.21×	4.49×
NM1	0.50	4.88×	5.15×	5.01×	5.25×	5.28×	5.76×	6.28×
RG1	6.86	1.45×	1.70×	1.70×	1.70×	1.69×	1.63×	1.72×
RG2	7.18	3.69×	4.23×	4.32×	4.28×	4.30×	4.32×	4.51×
RG3	5.91	5.46×	6.08×	6.82×	6.11×	6.12×	6.99×	7.31×
SC1	9.71	3.46×	3.71×	3.70×	3.73×	3.73×	3.85×	4.73×
XA1	2.00	4.37×	5.01×	4.94×	5.04×	5.03×	4.88×	5.39×
CT_BRAIN	0.50	3.08×	3.06×	2.89×	2.77×	2.47×	3.39×	4.28×
CT_ANKLE	0.50	9.30×	9.48×	5.02×	9.30×	9.14×	4.97×	6.87×
CT_ORT	0.50	2.36×	2.68×	2.38×	2.51×	2.26×	2.56×	3.00×
CT_CHEST	0.39	2.46×	2.82×	2.10×	2.52×	2.21×	2.23×	3.22×
MR_HEAD	0.13	2.11×	2.61×	2.65×	2.57×	2.53×	2.67×	2.86×
MR_INTERA	2.00	4.71×	3.68×	4.05×	3.75×	3.78×	5.04×	5.08×
DX_HAND	3.98	1.99×	2.24×	2.35×	2.26×	2.25×	2.37×	2.48×
DX_CHEST	3.84	1.43×	1.66×	1.82×	1.65×	1.63×	1.63×	1.70×
CR_THORAX	7.43	1.61×	1.82×	1.81×	1.84×	1.83×	1.81×	1.90×
PET1	0.13	2.37×	2.74×	2.81×	2.64×	2.52×	3.02×	3.21×
PET2	0.13	3.39×	3.39×	3.48×	3.16×	2.58×	4.37×	4.74×
CT_LUNG	0.50	2.40×	2.73×	2.45×	2.50×	2.25×	2.67×	3.15×
CT_PANCREAS	0.50	2.09×	2.36×	1.76×	2.18×	1.98×	1.85×	2.70×
MR_BRAIN	0.16	6.17×	7.27×	6.75×	7.31×	7.24×	7.62×	8.46×
MR_BREAST	0.13	3.31×	4.01×	3.94×	4.00×	3.91×	4.10×	4.48×
MR_PROSTATE	0.20	2.04×	2.30×	2.34×	2.28×	2.26×	2.49×	2.65×
PET_PSMA	0.08	11.88×	10.12×	7.67×	1.20×	1.18×	13.91×	15.78×
PET_LUNG	0.08	4.56×	2.97×	2.97×	1.00×	1.03×	5.21×	5.62×
Geomean	—	3.06×	3.36×	3.21×	3.04×	2.95×	3.42×	3.84×

VII. DISCUSSION

A. Reading the results together

Three observations tie the numbers together. First, MIC’s **ratio advantage over Zstandard** (about 10% in geometric mean, higher on 34 of 39 images) comes mainly from working on 16-bit residuals directly instead of splitting them into bytes, and that same choice removes the per-byte branches that slow decoding. Second, on the largest mammography images on AMD64, **HTJ2K’s block coder** regains the speed lead by amortizing its setup over many pixels; MIC stays within 0–20% there. Third, **multi-state decoding** is what makes MIC’s decoder competitive with HTJ2K’s vectorized block coder, and it costs nothing in file size.

B. Choosing a codec

For **low-latency viewing** (PACS, thumbnails, tomosynthesis playback), single-threaded MIC is a sensible default (~600–900 MB/s per request), with strip-parallel MIC for one large image at a time. For **smallest archival files**, JPEG-LS keeps a ~14% better ratio, at the cost of decoding 1.6–5.7× slower on ARM64. Where **DICOM transfer-syntax compatibility** is required, HTJ2K is the appropriate choice, since it is a standard and is particularly strong on large mammography images on AMD64. Where **no native code is permitted**

(browsers, sandboxes), the pure-Go reference or the 20 KB JavaScript decoder runs at about half the C decoder’s speed.

A more structured comparison—a Pareto view of ratio vs. decode speed [24], [25], plus a weighted decision matrix over ratio, speed, platform coverage, and deployment ease [26], [27], following deployment taxonomies for DICOM compression [18], [28]—puts MIC, HTJ2K, and JPEG-LS on the efficiency frontier (MIC for throughput, JPEG-LS for ratio, HTJ2K in between), with Delta + Zstandard dominated by MIC on both axes. HTJ2K remains the choice whenever standard compatibility is a hard requirement that the scoring does not capture. Table XV reports the weighted matrix.

C. Limitations

Small dataset. 39 images across eleven modalities is varied but far from a clinical archive; broader validation is needed. **Tuning on the test set.** The adaptive table-size thresholds and the predictor choice were informed by this dataset; the rules are simple (two thresholds, one predictor), which limits overfitting risk, but larger studies should confirm they generalize. **Baseline coverage.** Stronger byte-oriented baselines (Delta + byte/bit-shuffle + Zstd, DICOM RLE) would more sharply isolate the value of the 16-bit entropy stage; left to future work. **Reporting.** Tables give single representative throughput

TABLE IX
SINGLE-THREADED ENCODING THROUGHPUT (MB/S). MIC IS THE EIGHT-STATE C ENCODER. BOLD MARKS THE FASTEST CODEC PER ROW ON EACH PLATFORM.

Image	ARM64				AMD64			
	MIC	Δ +Zstd	HTJ2K	JPEG-LS	MIC	Δ +Zstd	HTJ2K	JPEG-LS
MR	457	7	236	91	344	5	246	62
CT	476	7	183	126	280	5	207	97
CR	525	2	186	110	322	2	286	75
XR	661	3	229	113	400	4	256	84
MG1	1,057	11	414	277	532	8	632	211
MG2	1,047	12	407	284	552	8	629	209
MG3	639	2	211	132	352	2	245	104
MG4	883	5	342	196	457	6	408	151
CT1	615	9	243	176	321	7	271	125
CT2	536	7	204	172	285	5	271	124
MG-N	639	2	214	140	358	2	248	104
MR1	616	9	205	114	367	6	241	86
MR2	685	7	216	118	409	4	309	85
MR3	768	19	264	167	441	12	354	123
MR4	585	7	212	163	367	5	312	142
NM1	633	6	208	162	340	5	313	115
RG1	595	4	218	84	396	5	252	66
RG2	737	4	263	151	425	3	360	100
RG3	757	6	276	183	389	4	412	126
SC1	667	5	225	203	416	4	306	169
XA1	612	5	203	133	356	4	303	98
CT_BRAIN	352	5	168	147	239	4	253	112
CT_ANKLE	955	15	290	327	450	11	358	261
CT_ORT	644	10	198	138	350	7	269	112
CT_CHEST	463	6	177	133	267	4	234	100
MR_HEAD	430	6	211	106	288	4	252	65
MR_INTERA	415	4	180	115	269	3	268	89
DX_HAND	587	4	218	102	378	3	261	65
DX_CHEST	636	6	209	109	389	6	241	77
CR_THORAX	585	4	226	91	397	4	261	68
PET1	479	7	240	129	295	5	252	111
PET2	550	6	236	145	269	4	256	103
CT_LUNG	586	10	195	141	337	7	258	110
CT_PANCREAS	454	7	193	130	292	5	240	98
MR_BRAIN	607	7	229	191	338	6	351	141
MR_BREAST	522	5	241	141	292	4	290	96
MR_PROSTATE	465	8	195	97	313	6	250	66
PET_PSMa	434	14	423	426	296	10	377	364
PET_LUNG	232	6	245	235	144	5	260	167
Geomean	582	6	231	148	343	5	293	110

values rather than full confidence intervals. **Not yet a DICOM transfer syntax.** MIC works on the extracted pixel array; real PACS integration would need a private encapsulation or a standardization step.

VIII. CONCLUSION

MIC is a 16-bit-native lossless codec for medical images: spatial prediction, 16-bit run-length coding, a large-alphabet table-based entropy coder, and a multi-state decoder for speed. On 39 DICOM images it reaches a geometric-mean ratio of $3.36\times$ (within $\sim 2\%$ of HTJ2K, $\sim 88\%$ of JPEG-LS) while being the fastest decoder on all 39 images on ARM64 and 36/39 on AMD64, and the fastest encoder on 38/39 ARM64 images and 32/39 on AMD64. A 20 KB JavaScript decoder and a WebAssembly build show the format can be decoded client-side in a browser. Future work: larger multi-hospital datasets; stronger shuffle-based baselines; full statistical reporting; an ARM64 vector kernel for the wavelet variant; and benchmarking the JavaScript decoder in real browsers. MIC is open source: <https://github.com/pappuks/medical-image-codec>.

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TABLE X

SINGLE-THREADED DECOMPRESSION THROUGHPUT (MB/S). MIC IS THE EIGHT-STATE C DECODER (AVX-512 ACCELERATION OF THE RLE AND DELTA INVERSE STAGES ON AMD64). BOLD MARKS THE FASTEST CODEC PER ROW ON EACH PLATFORM.

Image	ARM64				AMD64			
	MIC	Δ +Zstd	HTJ2K	JPEG-LS	MIC	Δ +Zstd	HTJ2K	JPEG-LS
MR	649	389	328	137	614	360	453	91
CT	495	406	299	158	444	393	329	110
CR	580	527	316	181	570	479	445	124
XR	668	512	324	131	626	509	393	88
MG1	762	685	612	481	916	596	913	336
MG2	753	692	622	492	942	599	902	334
MG3	600	433	314	175	547	362	392	118
MG4	788	504	508	246	686	469	634	158
CT1	600	450	357	224	535	432	389	142
CT2	570	471	342	212	522	458	383	131
MG-N	600	408	289	180	549	357	398	118
MR1	734	365	342	149	586	381	392	93
MR2	700	445	363	211	682	460	486	131
MR3	839	523	432	289	742	497	520	192
MR4	692	466	326	220	664	496	433	152
NM1	778	499	384	260	597	515	485	171
RG1	469	383	318	123	547	340	363	80
RG2	652	510	383	225	716	503	544	154
RG3	685	642	432	294	719	564	641	195
SC1	699	511	366	263	673	483	457	184
XA1	662	501	346	244	670	541	501	165
CT_BRAIN	528	466	326	190	420	477	354	122
CT_ANKLE	853	587	452	426	774	562	537	282
CT_ORT	538	442	348	190	565	430	384	124
CT_CHEST	500	445	297	178	446	428	333	110
MR_HEAD	595	372	280	167	679	371	406	100
MR_INTERA	590	563	304	178	454	527	436	133
DX_HAND	500	414	316	147	570	416	362	101
DX_CHEST	476	352	307	121	547	349	358	83
CR_THORAX	552	403	346	129	561	365	365	84
PET1	732	416	377	197	679	384	389	120
PET2	656	515	381	227	487	438	389	141
CT_LUNG	522	316	348	189	534	420	383	122
CT_PANCREAS	517	451	316	172	439	390	337	110
MR_BRAIN	866	567	422	375	708	514	641	221
MR_BREAST	640	545	350	221	759	449	510	156
MR_PROSTATE	679	422	353	165	545	386	408	103
PET_PSMA	759	678	475	684	747	655	539	395
PET_LUNG	502	498	283	278	365	508	363	170
Geomean	631	473	359	215	598	452	445	140

TABLE XI

FSE DECOMPRESSION THROUGHPUT (MB/S) FOR REPRESENTATIVE MODALITIES, SINGLE-THREAD PURE-GO DECODER. INCREASING THE NUMBER OF INDEPENDENT STATE CHAINS EXPOSES MORE INSTRUCTION-LEVEL PARALLELISM TO THE OUT-OF-ORDER ENGINE. PER-IMAGE ROWS ARE TEN REPRESENTATIVE MODALITIES; BOTH GEOMETRIC MEANS ARE OVER ALL 39 IMAGES.

Image	ARM64 (MB/s)			AMD64 (MB/s)		
	1-st	4-st	8-st	1-st	4-st	8-st
MR	1,066	1,576	1,068	656	1,007	847
CT	1,001	1,549	2,142	972	1,034	1,362
CR	2,941	4,882	6,245	1,456	1,844	1,709
XR	3,597	6,081	6,030	1,745	2,341	2,014
MG1	3,558	4,865	4,885	2,216	1,487	1,975
MG-N	3,949	6,647	6,037	1,940	2,545	2,415
NM1	1,971	2,661	2,570	917	1,346	1,356
RG1	1,768	4,356	5,332	1,239	1,709	1,869
DX_HAND	2,120	3,163	4,816	1,245	1,833	1,520
PET1	1,105	1,252	1,530	575	763	616
Geomean (39)	1,742	2,372	2,492	1,093	1,228	1,246

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TABLE XII

MULTI-THREADED DECOMPRESSION THROUGHPUT (MB/S) FOR MIC’S STRIP-PARALLEL MODE. THE “MIC” COLUMN REPRODUCES THE SINGLE-THREADED EIGHT-STATE BASELINE FROM TABLE X EXACTLY; N -STRIP COLUMNS DECODE THE SAME COMPRESSED BYTES IN N PARALLEL THREADS. BOTH ARM64 AND AMD64 COLUMNS COVER ALL 39 IMAGES.

Image	ARM64				AMD64			
	MIC	2-strip	4-strip	8-strip	MIC	2-strip	4-strip	8-strip
MR	649	698	939	1,033	614	281	369	291 [†]
CT	495	511	987	1,549	444	379	641	736
CR	580	865	1,771	3,227	570	768	1,404	2,236
XR	668	888	1,728	3,197	626	787	1,408	2,252
MG1	762	1,415	2,403	4,342	916	1,330	1,924	3,575
MG2	753	1,369	2,226	4,109	942	1,302	1,698	3,011
MG3	600	934	1,763	3,158	547	868	1,597	2,781
MG4	788	1,263	2,185	4,120	686	1,258	1,957	3,107
CT1	600	779	1,178	1,639	535	477	743	797
CT2	570	773	1,249	1,296	522	475	718	806
MG-N	600	988	1,852	3,569	549	871	1,644	2,913
MR1	734	931	1,375	1,898	586	517	729	827
MR2	700	1,003	1,699	2,750	682	786	1,064	1,476
MR3	839	919	1,381	2,035	742	627	861	853
MR4	692	954	1,301	2,063	664	573	813	877
NM1	778	964	1,410	1,868	597	561	777	771
RG1	469	586	1,068	1,964	547	626	1,113	1,833
RG2	652	1,095	1,887	3,269	716	922	1,523	2,494
RG3	685	1,081	2,027	3,289	719	991	1,596	2,504
SC1	699	1,171	2,005	3,174	673	1,042	1,690	2,598
XA1	662	973	1,692	2,404	670	806	1,170	1,443
CT_BRAIN	528	627	1,025	1,291	420	426	630	681
CT_ANKLE	853	1,082	1,745	1,713	774	654	869	877
CT_ORT	538	679	1,099	1,316	565	485	726	778
CT_CHEST	500	622	1,042	1,354	446	409	633	691
MR_HEAD	595	620	915	880	679	306	417	307
MR_INTERA	590	910	1,575	2,405	454	618	940	1,329
DX_HAND	500	605	1,113	2,051	570	619	1,133	1,697
DX_CHEST	476	566	1,003	1,971	547	592	990	1,712
CR_THORAX	552	588	1,128	1,733	561	624	1,141	1,937
PET1	732	637	854	854	679	289	406	317
PET2	656	669	776	923	487	341	356	320
CT_LUNG	522	689	1,107	1,462	534	468	752	808
CT_PANCREAS	517	506	1,018	1,316	439	390	652	688
MR_BRAIN	866	815	999	1,143	708	435	490	448
MR_BREAST	640	653	920	956	759	288	396	337
MR_PROSTATE	679	742	1,093	1,279	545	321	507	455
PET_PSMA [‡]	759	727	–	–	747	282	–	–
PET_LUNG [‡]	502	359	–	–	365	208	–	–

[†]The 130 KB MR image is too small to benefit from 8-way parallelism on AMD64: synchronization overhead exceeds the per-strip work, so 4-strip is faster than 8-strip. [‡]On the two 0.08 MB PET images, splitting into four or more strips drives the strip-compressed size to or above the single-stream size, so their 4- and 8-strip figures reflect near-raw copying rather than entropy decoding and are omitted.

TABLE XIII

JAVASCRIPT DECODER THROUGHPUT ON THE ARM64 REFERENCE (APPLE M4 PRO, NODEJS v22.16), FOR THE FOUR-STATE AND EIGHT-STATE FSE VARIANTS. MEDIAN OVER 20 ITERATIONS AFTER THREE WARM-UP RUNS.

Image	Dimensions	Ratio	4-state		8-state	
			ms	MB/s	ms	MB/s
MR [†]	256×256	2.35×	2.9	43	2.7	46
CT	512×512	2.24×	12.3	41	13.1	38
CR	2140×1760	3.69×	137.2	52	165.5	43
MG1	2457×1996	8.79×	70.4	133	80.0	117
MG3	4774×3064	2.29×	560.8	50	607.1	46
DX_HAND	1480×1410	2.24×	82.6	48	96.1	41
PET1	256×256	2.74×	2.1	60	2.5	50

[†]MR’s 0.13 MB image decodes in under 3 ms, at the JavaScript timing floor; the four-state advantage seen on the larger images is within run-to-run noise here.

TABLE XIV

PICS PARALLEL JAVASCRIPT DECODER (WORKER_THREADS) ON THE ARM64 REFERENCE, WITH FOUR-STATE AND EIGHT-STATE FSE PER STRIP. WORKER COUNT WAS SWEEPED OVER {1, 2, 4, 8, 14}; THE BEST WALL-TIME FOR EACH VARIANT IS REPORTED.

Image	Strips	4-state strips		8-state strips	
		ms	MB/s	ms	MB/s
MR	4	0.9	134	0.9	135
CT	4	4.1	123	4.5	112
CR	8	22.9	313	28.0	257
MG1	8	17.1	547	16.7	559
DX_HAND	8	17.1	233	17.8	224

TABLE XV

CODEC SELECTION MATRIX. EACH CRITERION IS NORMALISED SO THE BEST CODEC ON THAT ROW SCORES 1.00. QUANTITATIVE ROWS ARE GEOMETRIC MEANS ACROSS THE 39-IMAGE SET. BROWSER COMPATIBILITY: SHIPPED PURPOSE-BUILT JS OR WASM DECODER USED IN PRODUCTION = 1.00; WASM PORT OF THE NATIVE LIBRARY USED IN CLINICAL VIEWERS = 0.85; NO BROWSER PATH = 0.0.

DEPLOYMENT EASE: ZERO EXTERNAL RUNTIME DEPENDENCIES = 1.00; UBIQUITOUS SYSTEM DEPENDENCY = 0.90; C/C++ LIBRARY REQUIRING CMAKE BUILD AND LINK INTEGRATION = 0.70. WEIGHT COLUMNS w_P , w_A , w_B REFLECT TYPICAL PRIORITIES FOR (P) PACS VIEWER / LOW-LATENCY DECODE, (A) ARCHIVAL STORAGE, (B) BROWSER DELIVERY.

Criterion	w_P	w_A	w_B	MIC	HTJ2K	JPEG-LS
Compression ratio	0.20	0.50	0.10	0.88	0.89	1.00
Decompression throughput (ARM64)	0.30	0.10	0.15	1.00	0.57	0.34
Encoding throughput (ARM64)	0.05	0.10	0.05	1.00	0.40	0.25
AMD64 native	0.10	0.05	0.05	1.00	1.00	1.00
ARM64 native	0.10	0.05	0.05	1.00	1.00	1.00
Browser compatibility	0.05	0.00	0.40	1.00	0.85	0.85
Deployment ease	0.20	0.20	0.20	1.00	0.70	0.70
Weighted score (P) viewer	—	—	—	0.98	0.75	0.70
Weighted score (A) archival	—	—	—	0.94	0.78	0.80
Weighted score (B) browser	—	—	—	0.99	0.77	0.74